Rohit Patra

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Summary

Staff Research Scientist and former tenure-track faculty member combining ML research with direct product impact. I develop interpretable but rigorous policies, assessment metrics, and methods for uncertainty quantification that drive sustained online lifts across large-scale systems such as recommendation systems, personalized notifications, bandit platforms, and LLM+Graph agents. I have a strong record of peer-reviewed publications and measurable business results.

Work Experience

LinkedIn, New York, NY — August 2021 – Present

Staff ML Scientist, Applied Research

- *Economic Graph Reasoning Agent (LLM+Graph)* - core contributor to the design and training of LLM agent over LinkedIn's heterogeneous graph; evaluation design, training data pipelines, reward modelling, and metric definition.
- *Personalized Notifications (Decision Transformer RL)* — ML lead for a 10-engineer effort to deploy a Decision-Transformer + quantile-regression policy in a sequence-model-based, real-time notifications stack; owned training and inference; $+0.72\%$ sessions lift.
- *Contextual Bandit Platform* — Architected and implemented a new bandit platform with efficient uncertainty quantification (with coverage guarantees) compatible with billion-parameter LLMs; built continual-retraining, policy-safety gates, and interpretable evaluation metrics and authored company-wide playbooks; delivered $100M+ incremental revenue and $+0.75\%$ sessions lift; led 12 other engineers.
- *Data-Centric AI* — Proposed and built weak-supervision platform and covariate/concept-shift detection powering $10+$ models; reduced fake/abusive profiles by 15%; led 8 engineers.

University of Florida, Gainesville, FL

Tenure Track Assistant Professor, Department of Statistics — September 2016 – August 2021

- NSF funded collaborations across ML, Economics, Astronomy, and Statistics; published in Journal of the Royal Statistical Society Series B, JASA, AoS, Bernoulli, Biometrics; co-PI on NSF DMS-2210662 ($200,000).

Education

Columbia University, New York, NY

- PhD in Statistics, Advisor: Bodhisattva Sen — June 2016

Doctoral dissertation on using natural model constraints to do hyperparameter free inference and learning.

Indian Statistical Institute, Kolkata, India

- Bachelor's and Master's in Statistics — July 2010

Selected Publications and Preprints

- Generative Sequential Notification Optimization via Multi-Objective Decision Transformers *In Review at KDD* (2025+) (with B. Ocejo and others)
- BanditLP: Large-Scale Stochastic Optimization for Personalized Recommendations *The ACM Web Conference 2026* (2026) (with P. Nguyen and others)
- Efficient Uncertainty Quantification with Guarantees via Sketching *In Preparation* (2025+) (with G. Dexter and J. McInerney)
- Computationally efficient Laplace approximations for neural networks. *NeurIPS Workshop on Bayesian Decision-Making and Uncertainty* (2024). (with S. Raha, K. Khare)
- Least squares estimation of a monotone quasiconvex regression function. *Journal of the Royal Statistical Society Series B* (2024). (with S. Mukherjee, A. L. Johnson, H. Morita)
- A nonparametric approach for estimating consumer valuation distributions using second-price auctions. Revise and resubmit, *Annals of Applied Statistics* (2023). (with S. Mukherjee, K. Khare)

- Dimension reduction for integrative survival analysis. *Biometrics* (2023). (with A. Molstad)
- On least squares estimation under heteroscedastic and heavy tailed errors. *Annals of Statistics* (2022). (with A. K. Kuchibhotla)
- Least squares estimation in a single index model with convex Lipschitz link. *Journal of the American Statistical Association* (2021). (with A. K. Kuchibhotla, B. Sen)
- Efficient estimation in single index models through smoothing splines. *Bernoulli* (2020). (with A. K. Kuchibhotla)
- A consistent bootstrap procedure for the maximum score estimator. *Journal of Econometrics* (2018). (with E. Seijo, B. Sen)
- Estimation of a two-component mixture model with applications to multiple testing. *Journal of the Royal Statistical Society Series B* (2016). (with B. Sen)

Selected Grants and Awards

- *NSF grant, Division of Mathematical Sciences – 2210662* ($200,000)
  Title: Methodology for Qualitative Constraints in Semi-Parametric Models. Role: co-PI
- *National Fellowship in Basic Sciences* (Kishore Vaigyanik Protsahan Yojana), Department of Science and Technology, Government of India, 2005–2010.

Research Community Service

- Scientific Committee, IISA (2025); Co-organizer, UF Winter Statistics Workshop (2018, 2021); Co-organizer, Bayes Comp (2020).
- Graduate advising: committee member for 15+ theses.
- **Reviewer for**: JMLR, Journal of the Royal Statistical Society Series B, Journal of the American Statistical Association, AoS, AoAS, Econ Theory, Biometrika, and others.

Programming Skills

- PyTorch, TensorFlow, Python